

K-Medoids clustering

Medoid: A medoid is a point in the cluster from which the sum of distances to other data points is minimal.

Instead of using mean/centroid as the central point of the cluster, we use Medoid as the reference point in k-Medoids algorithm.

1. Initialization: Select k initial medoids randomly from the dataset.
2. Assignment Step: For each data point, calculate the distance to each medoid and assign to the cluster associated with the nearest medoid.

3. **Update Step:** For each cluster, select the data point that minimizes the sum of distances to all other points in the same cluster to be the new medoid.

↳ while the cost decreases: For each medoid m , for each data point o which is not a medoid in the same cluster:

> Swap m and o and compute the cost from o .

> If the cost from o is more than m , undo the swap

(Partitioning around clustering)

4. **Repeat until Convergence:** Repeat steps 2 & 3 until convergence

Medoids not changing }
Assignments not changing }
Max. number of iterations }

eg. Dataset:

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
A	1	2
B	1	4
C	1	0
D	10	2
E	10	4

Use k-medoids algorithm with $k=2$ with initial medoids as data points A and D. Use Euclidean distance as the distance metric.

Sol.	Distances				
	Feature 1 (x_1)	Feature 2 (x_2)	M1 (1,2)	M2 (10,2)	Assign ment
A	1	2	0	9	M1
B	1	4	2	9.2	M1
C	1	0	2	9.2	M1
D	10	2	9	0	M2
E	10	4	9.2	2	M2

$$M1: \{A, B, C\}$$

$$M2: \{D, E\}$$

$$A: 0 + 2 + 2 = 4$$

$$B: 2 + 0 + 4 = 6$$

$$C: 2 + 4 + 0 = 6$$

$$\text{New mediod :- } M_1 = A$$

$$\begin{array}{l} D: 0+2=2 \\ E: 2+0=2 \end{array} \left. \vphantom{\begin{array}{l} D \\ E \end{array}} \right\} \begin{array}{l} \text{select} \\ \text{any one} \end{array} \quad \text{New medoid} := M_2 = D$$

Medoids won't change after first iteration.

Q. Consider the following dataset:

Data Point	Feature 1 (x_1)	Feature 2 (x_2)
1	1	2
2	2	2
3	3	4
4	6	4
5	8	3

Assume that you run K-medoids algorithm with $k=2$ and Data points 1 and 3 as the initial medoids. using manhattan distance to compute distances

After the second assignment step, calculate the total sum of distances for all data points from their medoids.

Sol.	Data points		Distance		Assignment
			M ₁ (1,2)	M ₂ (3,4)	
①	1	2	①	4	M ₁
②	2	2	①	3	M ₁
③	3	4	4	①	M ₂
④	6	4	7	③	M ₂
⑤	8	3	8	⑥	M ₂

M₁: {1, 2} → Medoid ①

M₂: {3, 4, 5}

$$\text{Dis}(3): 0 + 3 + 6 = 9$$

$$\text{Dis}(4): 3 + 0 + 3 = ⑥$$

$$\text{Dis}(5): 6 + 0 + 3 = 9$$

↪ Medoid: ④

	Data points		Distance		Assignment
			M ₁ (1,2)	M ₂ (6,4)	
①	1	2	①	7	M ₁
②	2	2	①	6	M ₁
③	3	4	4	③	M ₂
④	6	4	7	①	M ₂
⑤	8	3	8	③	M ₂

Sum of distances

$$= 0 + 1 + 3 + 0 + 3$$

$$= \boxed{7}$$